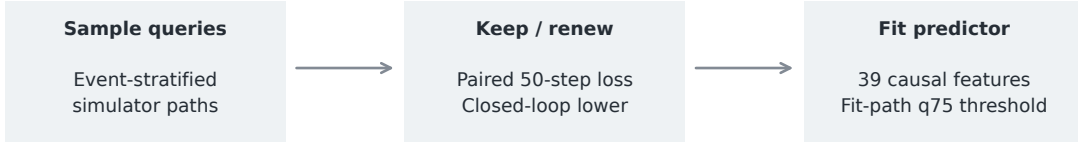
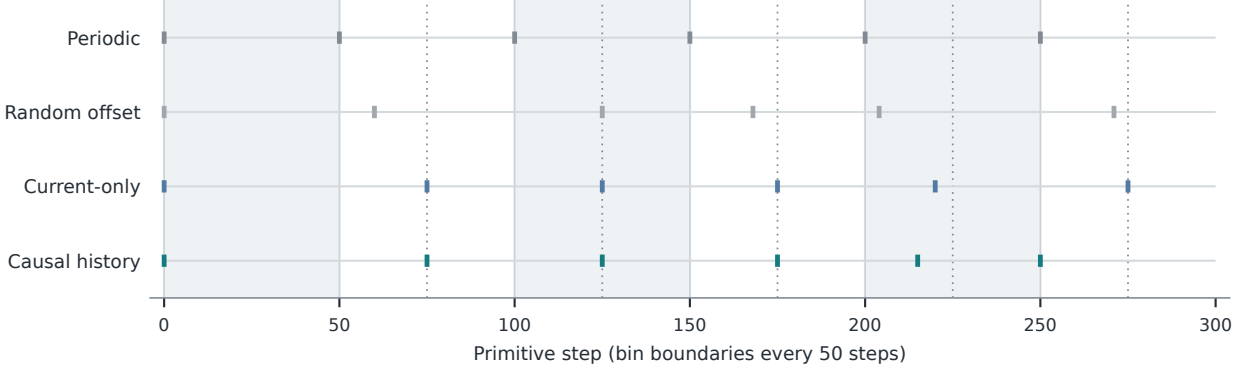
a **Paired supervision, then frozen deployment**



Offline: event metadata selects queries. Online: observable history only.

Check every 5 steps; renew at first score >= q, or at the offset-25 deadline.

b **Observed upper-call timing: first 300 steps**



Fixed example: optimizer root 209011, path 2180301; not selected for performance.

All modes: 24 calls/episode. Dotted lines: deadlines; lower actions continue at every step.